

Anomaly Detection Techniques using Deep Learning in IoT: A Survey

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Abstract—IoT technologies is improving life quality by enhancing several real-life smart applications. IoT includes large number of devices generating huge amount of data which needs large computation. Anomaly detection and security is the major concern in the IoT domain. This survey paper provides an overview of anomaly detection using machine learning and deep learning methods in IoT applications. Machine learning and deep learning are powerful tools for analyzing normal and abnormal behavior of IoT components and devices. In this paper we outline key issues in research and challenges using deep anomaly detection techniques for resource constrained devices in real-world problems of IoT. Fog computing move the computation at the device or the edge to solve some of the issues of network security and delay in computation.

Keywords—Anomaly Detection, Machine Learning, Deep learning, Fog Computing, CNN, DNN

I. INTRODUCTION

IoT is going through the phase of rapid growth nowadays. Application specific systems designed for specific purposes requires less time for computation but it has limited flexibility. IoT demands application, product and service platforms which can capture, communicate, store, access and share the data from the physical world. IoT is connecting all the devices referred to as ‘Things’ with internet and it has been growing trend in decades. IoT deals with sensors/actuators, RFID tags and communication technologies. RFID technology is transmitting the identification information from the microchip to the reader through wireless communication. Other IoT devices and technology such as barcode, smart phones and cloud computing are used to form network. For information exchange, processing and communication in IoT technical standards specification needs to be designed for high quality services. Standardization of the technologies will lead to success of IoT[1][2][3]. Layers of IoT are perception layer, network layer and business layer.

Machine learning and deep learning techniques have been used extensively for a many task including classification, regression and in IoT application areas such as intrusion detection system, computer vision, recommender system. Deep learning facilitates the analysis on fast and real data streams to extract features and predict future in IoT domain. DL is considered better than traditional machine learning as some features that might not be viewed to a human can be easily extracted and thus improve the accuracy [4]. In this paper review of different machine learning and deep learning techniques for anomaly detection are described for the IoT applications. Fog computing move the

computation at the device or the edge to solve some of the issues of network security and delay in computation.

II. ANOMALY DETECTION IN IOT

A. Anomalies and sources of anomalies

In real-world data-sets there are instances which are dissimilar to all others instances and are known as anomalies. The anomaly detection is to detect such patterns whose behavior is considered abnormal as compared to normal nodes. Different sources of anomalies are Intrusion detection system, Fraud detection and Data leakage. Anomaly detection is used in variety of areas in IoT domain as shown in Table I.

TABLE I. AREAS OF ANOMALY

Areas	Anomaly Description	Benefits
Smart Cities	Water leakage	Prevent water waste
	Light bulb broken	Maintenance time reduced
	Electricity leakage	To save energy
	Gas leakage	To save fuel
	Health detection	Life saving
Network Security	Intrusion detection, fraud detection, DDoS attacks	To secure data
Industries	Surface inspection of device	To remove defect on device

Intrusion detection: IoT devices are connected to the internet and remains vulnerable to security-related attacks. Heavy damage to IoT network is caused by attacks such as Denial-of-service (DoS) attacks and distributed denial-of-service (DDoS) attacks. Detection and prevention from those attacks is the major issue in IoT applications.

Fraud detection: IoT networks remain susceptible to stealing credit card information, bank account details, or other sensitive information during logins or online payments.

Data Leakage: Sensitive information from databases, file servers and other information sources can leak to any external entity which results in not only the loss of information, but also creates threat that can destroy confidential information from the system. Proper encryption mechanisms can prevent such leaks.

B. Types of Anomaly

It can be detected based on its type like Point-wise, Contextual or Collective [6].

Point-wise anomalies is to detect the points which deviate largely from the rest of the data points and used when the evolution of the series is not predictable. It is generally used for fraud detection.

Collective anomalies are detected by typical time series patterns such as recurring patterns or shapes from multiple IoT devices. In supply chain delay shipment is very common but if numerous delays then it needs investigation and thus collective analysis.

Contextual anomalies are detected when previous type of information or context is taken into account such as day of the week. Contexts are almost always very domain specific.

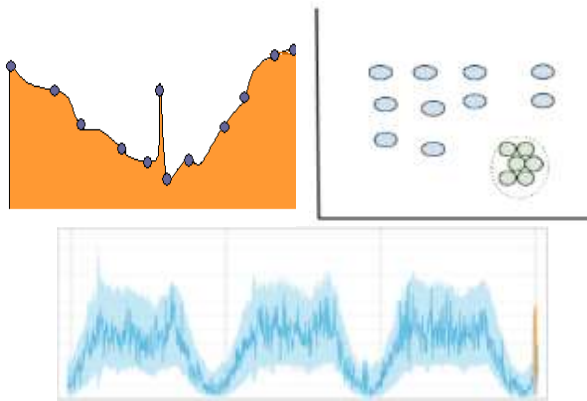


Fig. 1. Pointwise, Collective and Contextual Anomaly Types [6]

III. LITERATURE REIWEV

Machine learning and deep learning is widely used in anomaly detection in IoT applications. One of the most important tasks in manufacturing industries is the surface inspection. Deep convolution neural network method is applied to find the defects within the cut-out region which is detecting an anomaly within an anomaly image [7].

In multivariant time series anomaly detection and diagnosis abnormal status is detected after certain time steps and its causes is found. Multi-Scale Convolutional Recurrent Encoder-Decoder (MSCRED) [8] constructs multi-scale signature matrices in different time steps which is feed to a convolutional encoder and convolutional Long-Short Term Memory network is developed. Convolution decoder reconstruct the input signature matrices and then residual signature matrices are used to detect and diagnose the anomalies.

In health monitoring system real time anomaly is detected to monitor patient health. Deep neural network is applied to online physical activity monitoring dataset called MobiAct to detect real time fall detection of patient health based on accelerometer data giving 98.75 percent accuracy [9]. Ukil et al. [12] proposed a model in which smartphone is used for cardiac anomaly detection.

N.K et al. [11] proposed hybrid anomaly detection system that detect anomaly between router and nodes. Nodes temporally ban the communication for the prevention against anomaly. The developed system is test-bed on Zolertia and

Pandaboard based gateway and simulated on Contiki Cooja environment.

Rm et al. [13] proposed an improved optimization approach when the number of clusters are known and the weight is assigned to each data point in the cluster and then determine the relative position to the entire dataset. Experimental result on three different datasets shows that the improved optimized technique works better than K-Means.

Deep learning algorithm is used to develop an intelligent intrusion-detection system which detects malicious traffic in the network [14]. Resource constrained IoT devices having low power uses sequential deep neural-network models for real-time anomaly detection. Proposed IDS is deployed for IoT networks using a Raspberry Pi. DNN is applied using Keras, written in Python and tested using the Cooja network simulator developed in Contiki operating system.

Online skull stroke detection is implemented using CNN for feature extraction and then different classification methods bayesian method, random forest, multilayer perceptron, k nearest neighbor and support vector machine are applied[24]. Author performed image processing in python language and used tensor flow and keras library to build and trained the model with CNN and resultant data is then trained with SVM or MLP.

Deep migration learning model is introduced which enables the learning between different domains or tasks[25]. KDD CUP 99 is used as input where 41 number of input nodes and 5 output nodes are taken and after experiments it is found that 15 hidden layers provides the better performance. Evaluation of proposed model is done by conducting experiment in different scenarios. Data is collected on ubuntu platform imported into Microsoft windows 10 and simulated using MATLAB.

IV. FOG COMPUTING AND ANOMALY DETECTION

In IoT domain Fog and edge computing provide greater quality of service by getting faster response. [22]. Large amount of data generated by different IoT devices is processed near edge in fog and edge computing instead of transmitting it to cloud which provides the services with faster response and greater quality. Fog computing Characteristics are low latency, location awareness and wired or wireless access. Machine Learning and deep learning algorithms for Fog and edge computing are focused on techniques can process large amount of data quickly for the resource constrained devices.

A distributed model is proposed in which provides parallel training and parameters sharing by distributed fog nodes and detect network attack [23]. Experiment shows that distributed model is better than centralized model using deep learning approach however deep learning consumes more training time than shallow learning. Distributed attack detection provides increase in accuracy and avoids overfitting. Distributed detection scheme based on extreme learning machine is proposed for the resources available in cloud for time consuming and computationally expensive training tasks [26]. In edge cloud computing model distributed computing is combined with modern machine learning technology. Raw data is anonymized at edge and is then used for training ELM using cloud storage and computational facilities and suggest the detection model used at the edge for classification to detect threats and anomalies. Author used CTU database for IoT data

sources. The data source includes different types of attacks including DoS, the system is trained on one dataset and tested on another datasets. Experiments show that concatenating time windows of variable length increases effectiveness and addition of computing resources decreases the time required to train the ELM classifier.

Rare events are detected using IRESE deployed on edge device which continuously process the data and apply machine learning techniques [27]. Data collected by IoT devices is buffered for few seconds and then send to data framing where it is divided into smaller frames for feature extraction techniques and then rare event detection strategy is applied in two stages. At the first stage BIRCH Micro clustering technique is applied at the edge device to cluster high-speed incoming data in real time. At second stage Agglomerative macro clustering is applied to get macro clusters from the clusters formed at first stage based on Euclidean distance between clusters which finally divides into rare event and normal cluster. Experiment is conducted on audio data which include gunshot, siren, glass break and scream and results show that optimal value of threshold is selected where the all three values precision, recall and F1 is highest altogether.

V. KEY ISSUES AND CHALLENGES IN ANOMALY DETECTION

- **Missing data points:** Due to external environment data is lost which is difficult to detect.
- **Data corruption:** External factors or device malfunctioning corrupts the data and makes it difficult to differentiate anomaly and corrupted data.
- **Encrypted data:** to detect anomaly on encrypted data is difficult.
- **Sensor fusion:** It is difficult to collect data from different sensors and then aggregate for the results.
- **Real time detection:** Real time anomaly detection involves high speed streaming data and thus require faster response.
- **Noisy data:** Electronic transmission generates noisy data and thus is required to remove that from edge computing devices in IoT before forwarding it to cloud.
- **Traffic surge:** Huge amount of data can overload anomaly detection task.
- **Multivariate data:** Frequent change in data should also be considered.

TABLE II. SUMMARY OF DEEP LEARNING ALGORITHMS IN ANOMALY DETECTION

Author Name	Method Used	Methodology	Areas
Gruber, D. P., & Tabatabai. (2019).[7]	Unsupervised learning deep convolution neural network	Surface inspection in manufacturing industries. Image detection for anomaly.	Manufacturing Industries
] Zhang & N. V(2018)[8]	Multi-Scale Convolutional Recurrent Encoder-Decoder for time series multivariate	In time series multivariate signature matrices are used to detect and diagnose anomalies.	Intrusion Detection System
Mahfuz, (2019)[9]	Deep Neural network	Deep neural network is applied to online physical activity monitoring dataset called MobiAct to detect real time fall detection of patient health based on accelerometer data giving 98.75 percent accuracy	Health Monitoring System
Thanigaivelan, & Isoaho, J. (2018).[11]	Hybrid anomaly detection.	Detect attacks on nodes and react against anomaly by temporary communication ban on it	Intrusion detection system
A. Ukil & Puri , A. Pal(2016)[12]	Cardiac anomaly detection through a smartphone	Detection of anomalies in healthcare analytics based on IoT.	Health Care
Alguliyev (2017).[13]	An improved optimization approach for clusters	Improved optimization approach when the number of clusters are known and the weight is assigned to each data point in the cluster and then determine the relative position to the entire dataset.	Optimization Technique
Thamilarasu, G., & Chawla, S. (2019).[14]	Sequential deep neural-network	DNN is applied using Keras, written in Python and tested using the Cooja network simulator developed in Contiki operating system	Intrusion Detection System
Diro, A. A., & Chilamkurti, N. (2018). [23]	Distributed Deep learning	Distributed deep learning is applied for attack detection. Distributed model performs better than centralized model. Deep learning is one input layer and 3 hidden layers with softmax layer as output layer.	Intrusion detection system
Dourado, C. M. J. M., Pires, S. & H. C. De. (2019)[24]	CNN with classification methods	Author performed image processing in python language and used tensor flow and keras library to build and trained the model with CNN and resultant data is then trained with SVM or MLP	Online Stroke detection in Skull
Li, D., Deng, L., Lee, M., & Wang, H. (2019)[25]	Deep migration learning	Proposed model is evaluated by conducting experiment in different scenarios. Data is collected on ubuntu platform imported into Microsoft windows 10 and simulated using Matlab.	Feature extraction and intrusion detection system
Comput, J. P. D, & Palmieri, F. (2018).[26]	Extreme learning Machine	Extreme learning machine is proposed for distributed detection scheme for the resources which perform time consuming and computationally expensive training task available in cloud.	Intrusion detection system in edge computing

Haider, Z., Vecchio, M., Antonini, M., & Antonelli, F. (2019)[27]	IRESE unsupervised learning	At the first stage BIRCH Micro clustering technique is applied at the edge device to cluster high-speed incoming data in real time. At second stage Agglomerative macro clustering is applied to get macro clusters from the clusters formed at first stage based on Euclidean distance between clusters which finally divides into rare event and normal cluster.	Audio data detection in edge computing
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VI. CONCLUSIONS

In this paper several deep learning and machine learning techniques are described for the anomaly detection over the past few years in IoT. This review provides new insight to the researchers in the field of anomaly detection in IoT. Everyday a new unknown attack is discovered and thus there is need of new algorithms to detect the attack. IoT includes large number of devices and generate huge amount of data and thus large computation is needed. There is a need of feature extraction to faster the speed of computation. Anomaly detection at edge devices will provide faster response and greater quality of service in IoT. Real time data streaming in IoT needs real time anomaly detection. It is challenging and costly to detect anomaly using deep learning which requires high computing on resource constrained IoT devices.

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